

Combining Boundary and Interior Information in Ordered Multi-Threshold Regression Discontinuity Designs

Vahid Moghani¹ and Saeed Badri²

¹Erasmus School of Economics, Tinbergen Institute, and Erasmus
Centre for Health Economics Rotterdam

²Vrije Universiteit Amsterdam, Erasmus School of Economics, and Tinbergen Institute

April 25, 2026

Abstract

Ordered multi-threshold regression discontinuity designs often require aggregating local discontinuities across several known cutoffs. This paper shows that such aggregate parameters can be decomposed into boundary terms and smooth contrasts between adjacent thresholds. Interior observations do not identify a single RD jump, but they are informative about these smooth segment contrasts. We develop an integrated-derivative estimator for segment contrasts by numerically integrating nonparametric estimates of the derivative of the conditional mean between adjacent thresholds. We derive its asymptotic distribution, including the leading bias and a feasible variance formula. The main result shows that the integrated-derivative estimator is asymptotically independent of the corresponding boundary-level estimator for the same smooth contrast. This independence yields

¹We thank Chen Zhou, Guido Imbens, Stefan Wager, Peter Hull, and Jonathan Roth for their constructive comments.

a transparent inverse-variance combination of boundary and interior signals. The combined estimator is an augmentation of boundary-level estimation and is most useful when boundary information is sparse or noisy. Simulations confirm the distributional approximation and show modest gains under regular designs and larger gains in sparse-boundary designs.

1 Introduction

Regression discontinuity (RD) designs identify causal effects from discontinuities in conditional mean outcomes at known cutoffs (Hahn et al., 2001). In many applications, the running variable crosses more than one cutoff. Examples include age-based eligibility rules, school-entry rules repeated across cohorts, benefit schedules with several thresholds, and institutional rules that assign treatment intensity according to ordered score intervals. In such settings, researchers often summarize the evidence by estimating a weighted average of local discontinuities. Existing approaches typically estimate these effects by recentering observations around each cutoff, pooling the normalized running variable, or combining cutoff-specific estimates (Cattaneo et al., 2016; Bertanha, 2020). These methods are boundary-based: they use observations close to each cutoff to estimate the corresponding left and right limits of the conditional mean.

This paper studies a complementary source of information that arises in ordered multi-threshold designs. Let $t_1 < \dots < t_K$ denote known thresholds and let $g(z) = \mathbb{E}[Y \mid Z = z]$ be smooth on each open interval between adjacent thresholds, while allowing jumps at the thresholds. A weighted average of local discontinuities,

$$\tau = \sum_{k=1}^K \omega_k \{g(t_k^+) - g(t_k^-)\},$$

can be rewritten as a combination of outer endpoint terms and smooth contrasts between adjacent thresholds. These between-threshold contrasts are not RD jumps. They compare the right limit after one threshold, $g(t_{k-1}^+)$, with the left limit before the next threshold, $g(t_k^-)$. Under smoothness of g on (t_{k-1}, t_k) , the unweighted segment contrast satisfies

$$g(t_k^-) - g(t_{k-1}^+) = \int_{t_{k-1}}^{t_k} g'(z) dz.$$

Interior observations therefore do not identify a single local discontinuity, but

they do provide information about the smooth segment contrasts that enter aggregate multi-threshold estimands.

We use this decomposition to construct an interior estimator for each smooth segment contrast. For an adjacent-threshold segment (t_{k-1}, t_k) , define $\Delta_k = t_k - t_{k-1}$ and normalize $x = (z - t_{k-1})/\Delta_k \in [0, 1]$. Let

$$m_k(x) = g(t_{k-1} + x\Delta_k),$$

with endpoint limits inherited from the original scale, so that $m_k(0^+) = g(t_{k-1}^+)$ and $m_k(1^-) = g(t_k^-)$. The unweighted segment contrast is

$$m_k(1^-) - m_k(0^+) = \int_0^1 m'_k(u) du.$$

The proposed estimator partitions the normalized segment into bins, estimates $m'_k(u)$ at bin midpoints using local polynomial regression, and aggregates these derivative estimates by numerical integration. We derive its leading bias and asymptotic variance and establish a feasible central limit theorem.

The main contribution of the paper is to introduce an additional source of information for estimating the smooth segment contrasts in ordered multi-threshold designs. The integrated-derivative estimator provides a second signal for $m_k(1^-) - m_k(0^+)$ by exploiting variation inside the segment. We show that this interior estimator is asymptotically independent of the conventional boundary-level estimator that estimates $m_k(1^-)$ and $m_k(0^+)$ directly using one-sided local polynomial regressions at the segment endpoints. This independence follows from the geometry of the weighting schemes: the boundary-level estimator concentrates on shrinking neighborhoods of 0^+ and 1^- , while the integrated-derivative estimator distributes weight across the segment interior, so their effective overlap becomes asymptotically negligible.

This asymptotic independence gives a simple information-fusion rule. For each segment contrast, the boundary-level and integrated-derivative estimators can be combined using inverse-variance weights. The resulting estimator

combines two signals for the same smooth contrast: a boundary signal based on one-sided endpoint estimation and an interior signal based on integrated derivative estimation. The precision gain is design-dependent and is largest when boundary information is limited, for example when observations are sparse near adjacent thresholds or when outcome variance is high near the segment endpoints. The method is therefore an augmentation for ordered multi-threshold RD estimands, not a replacement for local estimation of a single RD jump.

The paper makes three contributions. First, it clarifies that aggregate estimands in ordered multi-threshold RD designs can contain smooth between-threshold contrasts in addition to local discontinuities. This distinction matters because interior observations are informative about the former but not about a single RD jump. Second, it develops an integrated-derivative estimator for these segment contrasts and derives its asymptotic distribution, including the leading bias and a feasible variance expression. Third, it proves asymptotic independence between the interior estimator and the conventional boundary-level estimator for the same segment contrast, which justifies an inverse-variance combination of the two signals.

The paper contributes to the literature on RD designs and, more specifically, to work on designs with multiple cutoffs. Hahn et al. (2001) formalized identification and estimation in RD designs using discontinuities in conditional expectations. Subsequent work has refined local polynomial estimation and inference at RD boundaries, including robust bias-corrected inference and coverage-error-optimal bandwidth selection (Calonico et al., 2014, 2020). For multi-cutoff settings, Cattaneo et al. (2016) show how pooled and cutoff-specific estimands should be interpreted, while Bertanha (2020) studies estimation of average treatment effects with many thresholds. Our contribution is different. We exploit the ordered structure of multiple thresholds to estimate the smooth segment contrasts that appear in aggregate parameters and show how these interior estimates can be combined with boundary-level estimates for the same contrasts.

The structure of the paper is as follows. Section 2 defines the ordered multi-threshold RD setting and derives the decomposition of aggregate estimands into boundary terms and smooth segment contrasts. Section 3 studies a single segment and develops boundary-level and integrated-derivative estimators for the same smooth contrast. Section 4 establishes the joint asymptotic behavior of the two estimators and derives the inverse-variance combination. Section 6 presents simulation evidence under regular and sparse-boundary designs. Section 7 concludes.

2 Ordered Multi-Threshold RD Estimands

Empirical research often uses RD designs in settings where decisions depend on more than one cutoff. Examples include school-entry rules repeated across cohorts, age-based eligibility rules, benefit schedules with several thresholds, and institutional rules that assign treatment intensity according to ordered score intervals. In these settings, the running variable crosses a sequence of known thresholds, and each threshold may induce a discontinuity in the conditional mean outcome.

Consider N independent observations (z_i, y_i) , where $z_i \in [a, b]$. Let

$$g(z) = \mathbb{E}[y_i \mid z_i = z]$$

denote the conditional mean outcome. Suppose there are known ordered thresholds

$$t_1 < t_2 < \cdots < t_K,$$

and that $g(\cdot)$ is smooth on each open interval between adjacent thresholds, while allowing jumps at the thresholds:

$$g(t_k^-) \neq g(t_k^+), \quad k = 1, \dots, K.$$

The local discontinuity at threshold t_k is

$$\tau_k = g(t_k^+) - g(t_k^-).$$

Researchers often summarize local discontinuities by a weighted average,

$$\tau = \sum_{k=1}^K \omega_k \tau_k, \quad \text{with } \sum_{k=1}^K \omega_k = 1.$$

The weights ω_k may be chosen by the researcher, set equal across thresholds, or induced by the empirical distribution of observations near the thresholds. Standard multi-cutoff RD approaches typically estimate such objects by recentering observations around each cutoff, pooling the normalized running variable, or combining cutoff-specific estimates (Cattaneo et al., 2016; Bertanha, 2020).

The ordered structure of the thresholds implies

$$\tau = \omega_K g(t_K^+) - \omega_1 g(t_1^-) - \sum_{k=2}^K [\omega_k g(t_k^-) - \omega_{k-1} g(t_{k-1}^+)].$$

Equivalently,

$$\tau = \omega_K g(t_K^+) - \omega_1 g(t_1^-) - \sum_{k=2}^K \delta_k, \tag{1}$$

where

$$\delta_k = \omega_k g(t_k^-) - \omega_{k-1} g(t_{k-1}^+).$$

The terms τ_k are RD jumps. In contrast, the terms δ_k are weighted between-threshold contrasts. When adjacent weights are equal, $\omega_k = \omega_{k-1}$, δ_k is proportional to the smooth segment contrast $g(t_k^-) - g(t_{k-1}^+)$. With unequal adjacent weights, δ_k also contains the difference in weights attached to the two endpoint levels. Section 5 treats this general weighted case explicitly. The key point is that, when $g(\cdot)$ is smooth on (t_{k-1}, t_k) , interior observations are informative about the smooth component of these between-threshold terms. They do not identify a single RD jump, but they can identify segment contrasts that enter

aggregate parameters such as (1).

For the main theoretical analysis, we study one such segment after normalization. On the segment $[t_{k-1}, t_k]$, let $\Delta_k = t_k - t_{k-1} > 0$ and define

$$x = (z - t_{k-1})/\Delta_k \in [0, 1], \quad m(x) = g(t_{k-1} + x\Delta_k).$$

Then

$$m(0^+) = g(t_{k-1}^+), \quad m(1^-) = g(t_k^-),$$

and

$$m'(x) = \Delta_k g'(t_{k-1} + x\Delta_k).$$

The unweighted segment contrast is

$$\delta = m(1^-) - m(0^+) = \int_0^1 m'(u) du. \quad (2)$$

The weighted case in (1) is considered in Section 5.

3 Boundary and Interior Estimation of a Segment Contrast

This section studies estimation of the smooth segment contrast δ in (2). We observe independent data $\{(X_i, Y_i)\}_{i=1}^N$ satisfying

$$Y_i = m(X_i) + \varepsilon_i, \quad \mathbb{E}[\varepsilon_i | X_i] = 0, \quad \text{Var}(\varepsilon_i | X_i) = \sigma^2(X_i),$$

where $X_i \in [0, 1]$. The target is the difference between the two one-sided endpoint limits,

$$\delta = m(1^-) - m(0^+).$$

There are two signals for this contrast. The boundary-level estimator estimates $m(0^+)$ and $m(1^-)$ directly using one-sided local polynomial regressions. The integrated-derivative estimator uses the identity $\delta = \int_0^1 m'(u) du$ and estimates

m' inside the segment.

3.1 Boundary-level estimator

The boundary-level estimator is

$$\hat{\delta}_{\text{levels}} = \hat{m}(1^-) - \hat{m}(0^+).$$

This is the standard endpoint estimator for a smooth contrast. It is analogous to the level-estimation step used in RD estimation, except that the target here is not a jump at a cutoff but a contrast between the right limit after one threshold and the left limit before the next threshold.

Let $\hat{m}(0^+)$ and $\hat{m}(1^-)$ denote the intercepts from one-sided degree- p local polynomial regressions at the left and right endpoints. For example, at the left endpoint,

$$\hat{\alpha}_0 = \arg \min_{\alpha} \sum_{i=1}^N \left\{ Y_i - \sum_{j=0}^p \alpha_j X_i^j \right\}^2 K(X_i/h_0) \mathbf{1}\{0 \leq X_i \leq h_0\}, \quad \hat{m}(0^+) = e_1^\top \hat{\alpha}_0,$$

where $e_1 = (1, 0, \dots, 0)^\top$. The right endpoint estimator is defined analogously using observations with $1 - h_1 \leq X_i \leq 1$, centered at 1. The selector is e_1 , not the derivative selector, because the boundary estimator targets endpoint levels.

From standard local polynomial theory (Fan and Yao, 2005), if $h_0 = c_0 h$, $h_1 = c_1 h$, and $h \rightarrow 0$, then

$$\sqrt{Nh} [\hat{\delta}_{\text{levels}} - \delta - \text{Bias}_{\text{levels}}] \xrightarrow{d} \mathcal{N} \left(0, \frac{R_1}{c_1} \frac{\sigma^2(1^-)}{f(1^-)} + \frac{R_0}{c_0} \frac{\sigma^2(0^+)}{f(0^+)} \right), \quad (3)$$

where R_0 and R_1 are the squared one-sided equivalent-kernel constants for the endpoint intercept estimators. The leading bias $\text{Bias}_{\text{levels}}$ is the usual one-sided local-polynomial endpoint bias of order h^{p+1} . In the simulations below, $p = 1$, and the endpoint intercepts are estimated by one-sided local linear regressions with the triangular kernel.

Under regular designs, the boundary-level estimator has the familiar $\sqrt{Nh}$ convergence rate. It is the benchmark estimator for the segment contrast. The integrated-derivative estimator introduced next provides an additional signal for the same contrast.

3.2 Integrated-derivative estimator

The integrated-derivative estimator uses

$$\delta = m(1^-) - m(0^+) = \int_0^1 m'(u) du.$$

Partition the interval $[0, 1]$ into B subintervals of length $2h$, with $h = 1/(2B)$, and let c_b denote the midpoint of bin b . At each midpoint c_b , estimate $m'(c_b)$ by local polynomial regression using observations in

$$I_b = [c_b - h, c_b + h].$$

Specifically,

$$\hat{\beta}(c_b) = \arg \min_{\beta} \sum_{X_i \in I_b} \left(Y_i - \sum_{j=0}^p \beta_j (X_i - c_b)^j \right)^2 K_h(X_i - c_b),$$

with $K_h(u) = K(u/h)/h$. The estimator of $m'(c_b)$ is

$$\hat{\beta}_1(c_b) = e_2^T \hat{\beta}(c_b) = \sum_{i=1}^N w_{i,b} Y_i, \quad (4)$$

where $w_{i,b} = w^T((X_i - c_b)/h)$, and

$$w^T(u) = e_2^T S_N^{-1} [1, uh, \dots, (uh)^p]^T \frac{K(u)}{h}.$$

The matrix S_N has (i, j) th element $S_{N,i+j}$, with

$$S_{N,j} = \sum_{i=1}^N K_h(X_i - c_b)(X_i - c_b)^j.$$

Aggregating the derivative estimates gives

$$\hat{\beta}_{\text{avg}} = 2h \sum_{b=1}^B \hat{\beta}_1(c_b) = \sum_{i=1}^N w_{N,i} Y_i,$$

where $w_{N,i}$ is the total weight assigned to observation i .

We impose the following assumptions.

Assumption 1 (Smoothness of m). $m \in C^{\max\{p+1, 5\}}([0, 1])$ for some integer $p \geq 1$.

Assumption 2 (Design Density). Each X_i takes values in $[0, 1]$ with a density f that is continuous, strictly positive on $[0, 1]$ (no atoms), and has a bounded second derivative.

Assumption 3 (Bounded Conditional Variance). $\sigma^2(x) = \text{Var}(\varepsilon_i \mid X_i = x)$ is continuous and bounded on $[0, 1]$.

Assumption 4 (Kernel Conditions). The kernel K is symmetric, bounded, supported on $[-1, 1]$, and satisfies

$$\int_{-1}^1 K(u) du = 1, \quad \int_{-1}^1 |u|^{p+1} |K(u)| du < \infty.$$

Define the equivalent kernel for the derivative estimator by

$$K^*(u) = e_2^T S^{-1}(1, u, \dots, u^p)^T K(u),$$

where S is a $(p+1) \times (p+1)$ matrix with entries

$$S_{i,j} = \mu_{i+j-2} \quad \text{and} \quad \mu_j = \int u^j K(u) du.$$

It follows that K^* is also supported and bounded on $[-1, 1]$.

Assumption 5 (Bandwidth). Let $h = h(N) \rightarrow 0$ as $N \rightarrow \infty$ and assume

$$\frac{Nh}{\log N} \rightarrow \infty, \quad Nh^2 \rightarrow \infty, \quad \sqrt{N}h^3 \rightarrow 0.$$

Assumption 6 (Higher-Order Moment of ε_i). There exists $\delta > 0$ and a constant $C < \infty$ such that

$$\mathbb{E}[|\varepsilon_i|^{2+\delta} \mid X_i = x] \leq C \quad \text{for all } x \in [0, 1],$$

uniformly in i .

4 Combining Boundary and Interior Signals

This section gives the main information-fusion result. The boundary-level estimator and the integrated-derivative estimator both target

$$\delta = m(1^-) - m(0^+).$$

They use different parts of the segment. Both estimators are valid estimators of the same smooth segment contrast. The distinction is not that the interior estimator identifies a different parameter, but that it extracts an additional signal for the same parameter from a different part of the support. The contribution of this section is to show that this interior signal is asymptotically independent of the boundary-level signal and can therefore be combined with it. The boundary-level estimator concentrates on shrinking neighborhoods of 0^+ and 1^- . The integrated-derivative estimator uses local derivative estimates across the segment. The central result is that the overlap between their effective weighting schemes is asymptotically negligible.

4.1 Asymptotic distribution of the integrated-derivative estimator

Theorem 1 (Integrated-derivative estimator for a segment contrast). *Let $\{(X_i, Y_i)\}_{i=1}^N$ be independent observations from the model*

$$Y_i = m(X_i) + \varepsilon_i, \quad \mathbb{E}[\varepsilon_i | X_i] = 0, \quad \text{Var}(\varepsilon_i | X_i) = \sigma^2(X_i),$$

where $X_i \in [0, 1]$ has density f . Suppose Assumptions 1–6 hold with polynomial degree $p \geq 2$, so that $m(\cdot)$ is $(p+1)$ -times continuously differentiable on $[0, 1]$, and we use a degree- p local polynomial estimator to estimate $m'(\cdot)$. Define

$$\Gamma = \int_0^1 m'(u) \, du = m(1^-) - m(0^+).$$

Partition $[0, 1]$ into B subintervals each of length $2h = 1/B$, with midpoints $c_b = (2b-1)h$ for $b = 1, \dots, B$. For each midpoint c_b , let $\hat{\beta}_1(c_b)$ be the local polynomial estimator of the derivative $m'(c_b)$. Define

$$\hat{\beta}_{\text{avg}} = 2h \sum_{b=1}^B \hat{\beta}_1(c_b).$$

Then, under Assumptions 1–6, and conditional on the design $X = \{X_1, \dots, X_N\}$,

$$\sqrt{N}h^2 \left(\hat{\beta}_{\text{avg}} - \Gamma - \text{Bias} \right) \xrightarrow{d} \mathcal{N} \left(0, 2 \int_{-1}^1 [K^*(u)]^2 \, du \int_0^1 \frac{\sigma^2(u)}{f(u)} \, du \right),$$

where

- If $p > 2$, then the local-polynomial bias is $o(h^2)$, and the dominant bias term comes from the midpoint rule:

$$\text{Bias} = -\frac{h^2}{6} \left[m''(1^-) - m''(0^+) \right].$$

- If $p = 2$, both the midpoint rule and the local-polynomial estimator contribute

to the bias at order h^2 :

$$\text{Bias} = \frac{h^2}{6} \left(\int_{-1}^1 u^3 K^*(u) du - 1 \right) [m''(1^-) - m''(0^+)].$$

Theorem 1 establishes that the interior estimator is asymptotically normal and has a feasible variance expression. Its variance depends on the density and conditional variance over the segment through

$$\int_0^1 \frac{\sigma^2(u)}{f(u)} du.$$

This differs from the boundary-level estimator in (3), whose asymptotic variance depends on the behavior of f and σ^2 at 0^+ and 1^- .

4.2 Asymptotic independence

Theorem 2 (Boundary–interior asymptotic independence). *Let $\hat{\beta}_{\text{avg}}$ be the integrated-derivative estimator with bandwidth h as in Theorem 1, and write*

$$\hat{\beta}_{\text{avg}} - \mathbb{E}[\hat{\beta}_{\text{avg}} | X] = \sum_{i=1}^N w_i \varepsilon_i, \quad \max_i |w_i| \leq \frac{C}{Nh}, \quad \sum_{i=1}^N w_i^2 = \Theta_p\left(\frac{1}{Nh^2}\right).$$

Let the boundary levels at 0 and 1 be estimated by one-sided degree- p local polynomials with asymmetric equivalent kernels and bandwidths $h_0 = c_0 h$ (at 0) and $h_1 = c_1 h$ (at 1), where $0 < c_{\min} \leq c_0, c_1 \leq c_{\max} < \infty$:

$$\hat{\delta}_{\text{levels}} = \hat{m}(1^-) - \hat{m}(0^+), \quad \hat{\delta}_{\text{levels}} - \mathbb{E}[\hat{\delta}_{\text{levels}} | X] = \sum_{i=1}^N v_i \varepsilon_i,$$

with

$$\max_i |v_i| \leq \frac{C}{Nh}, \quad \sum_{i=1}^N v_i^2 = \Theta_p\left(\frac{1}{Nh}\right),$$

and $v_i \neq 0$ only if $X_i \in [0, A_0 h] \cup [1 - A_1 h, 1]$, where $A_0, A_1 > 0$ depend only on the one-sided equivalent kernels and on c_0, c_1 .

Under Assumptions 1–6 (in particular $Nh \rightarrow \infty$ and $h \rightarrow 0$ and Assumption 2),

conditionally on X ,

$$\text{Cov}(\hat{\beta}_{\text{avg}}, \hat{\delta}_{\text{levels}} \mid X) = O_p\left(\frac{A_0 f(0+) + A_1 f(1-)}{Nh}\right),$$

and

$$\text{Corr}(\hat{\beta}_{\text{avg}}, \hat{\delta}_{\text{levels}} \mid X) = O_p\left((A_0 f(0+) + A_1 f(1-)) h^{1/2}\right) \rightarrow 0.$$

In particular, together with the marginal CLTs for $\hat{\beta}_{\text{avg}}$ and the standard boundary local-polynomial CLT for $\hat{\delta}_{\text{levels}}$, this implies that the jointly scaled vector is asymptotically bivariate normal with a diagonal covariance.

The proof is based on the covariance of two linear representations. Conditional on the design,

$$\hat{\beta}_{\text{avg}} - \mathbb{E}[\hat{\beta}_{\text{avg}} \mid X] = \sum_{i=1}^N w_i \varepsilon_i, \quad \hat{\delta}_{\text{levels}} - \mathbb{E}[\hat{\delta}_{\text{levels}} \mid X] = \sum_{i=1}^N v_i \varepsilon_i.$$

Thus,

$$\text{Cov}(\hat{\beta}_{\text{avg}}, \hat{\delta}_{\text{levels}} \mid X) = \sum_{i=1}^N w_i v_i \sigma^2(X_i).$$

The boundary weights v_i are nonzero only when X_i lies in endpoint neighborhoods of total length proportional to h . The number of observations in these neighborhoods is therefore $O_p(Nh)$. Since $\max_i |w_i| \leq C/(Nh)$ and $\max_i |v_i| \leq C/(Nh)$, the covariance is $O_p((Nh)^{-1})$. Dividing by the product of the standard deviations, of orders $(Nh^2)^{-1/2}$ and $(Nh)^{-1/2}$, gives a correlation of order $O_p(h^{1/2})$, which converges to zero.

4.3 Inverse-variance combination

Let

$$\hat{\delta}_{\text{int}} = \hat{\beta}_{\text{avg}}$$

denote the integrated-derivative estimator and let

$$\hat{\delta}_{\text{bnd}} = \hat{\delta}_{\text{levels}}$$

denote the boundary-level estimator. Let $\hat{V}_{\text{int}}$ and $\hat{V}_{\text{bnd}}$ denote feasible variance estimators for $\hat{\delta}_{\text{int}}$ and $\hat{\delta}_{\text{bnd}}$, respectively, on the original estimator scale. The inverse-variance combined estimator is

$$\hat{\delta}_{\text{comb}} = \frac{\hat{V}_{\text{bnd}}^{-1} \hat{\delta}_{\text{bnd}} + \hat{V}_{\text{int}}^{-1} \hat{\delta}_{\text{int}}}{\hat{V}_{\text{bnd}}^{-1} + \hat{V}_{\text{int}}^{-1}}.$$

Equivalently,

$$\hat{\delta}_{\text{comb}} = \lambda \hat{\delta}_{\text{bnd}} + (1 - \lambda) \hat{\delta}_{\text{int}}, \quad \lambda = \frac{\hat{V}_{\text{bnd}}^{-1}}{\hat{V}_{\text{bnd}}^{-1} + \hat{V}_{\text{int}}^{-1}}.$$

The asymptotic independence result justifies ignoring the covariance term in the leading variance approximation. A feasible variance estimator for the combined estimator is therefore

$$\hat{V}_{\text{comb}} = \left(\hat{V}_{\text{bnd}}^{-1} + \hat{V}_{\text{int}}^{-1} \right)^{-1}.$$

Under regular designs, the boundary estimator has the faster convergence rate and will typically receive most of the weight. The integrated-derivative estimator is most useful when boundary information is limited, for example when the density is low near 0^+ or 1^- , when the conditional variance is high near the segment endpoints, or when ordered multi-threshold designs produce thin local samples around some thresholds.

5 Weighted Segment Contrasts and Multi-Threshold Aggregation

The previous sections focused on the unweighted segment contrast

$$\delta = m(1^-) - m(0^+).$$

The decomposition in (1) contains weighted segment contrasts of the form

$$\delta_w = \omega_1 m(1^-) - \omega_0 m(0^+).$$

By the product rule $(\pi m)' = \pi' m + \pi m'$ and integration by parts, we have the representation

$$\delta_w = \int_0^1 \pi(u) m'(u) du + \int_0^1 \pi'(u) m(u) du, \quad (5)$$

for any function $\pi : [0, 1] \rightarrow \mathbb{R}$ with $\pi(1^-) = \omega_1$ and $\pi(0^+) = \omega_0$.

Let $\{I_b\}_{b=1}^B$ be the disjoint bins of width $2h$, with midpoints $c_b = (2b-1)h$. For each c_b , let $\hat{\beta}_1(c_b)$ be the degree- p local-polynomial estimator of $m'(c_b)$ from Theorem 1. Given a quadratic weight function $\pi(\cdot)$ with $\pi(1^-) = \omega_1$ and $\pi(0^+) = \omega_0$, define

$$\hat{\delta}_w := 2h \sum_{b=1}^B \pi(c_b) \hat{\beta}_1(c_b) + \int_0^1 \pi'(u) \hat{m}(u) du, \quad (6)$$

where $\hat{m}$ is any estimator satisfying

$$\int_0^1 q(u) \hat{m}(u) du = \int_0^1 q(u) m(u) du + O_p(N^{-1/2})$$

for each fixed quadratic q .

Theorem 3 (Weighted integrated-derivative estimator). *Suppose the assumptions of Theorem 1 hold (including the bandwidth condition) with local-polynomial degree $p \geq 2$. Let $\pi(\cdot)$ be a quadratic polynomial with $\pi(1^-) = \omega_1$, $\pi(0^+) = \omega_0$ and*

$\int_0^1 \pi'(u) \hat{m}(u) du = 0$. Then, conditional on X ,

$$\sqrt{Nh^2} \left(\hat{\delta}_w - \delta_w - \text{Bias}_\pi \right) \xrightarrow{d} \mathcal{N} \left(0, 2 \int_{-1}^1 K^{*2}(u) du \int_0^1 \frac{\pi^2(u) \sigma^2(u)}{f(u)} du \right),$$

where the bias satisfies

$$\text{Bias}_\pi = \begin{cases} -\frac{h^2}{6} \left[(\pi m')'(1^-) - (\pi m')'(0^+) \right] + o(h^2), & \text{if the local-poly degree} > 2, \\ \frac{h^2}{6} \left(\int_{-1}^1 u^3 K^*(u) du - 1 \right) \left[(\pi m')'(1^-) - (\pi m')'(0^+) \right] + o(h^2), & \text{if the degree} = 2, \end{cases}$$

with $(\pi m')'(u) = \pi'(u) m'(u) + \pi(u) m''(u)$.

For the aggregate RD parameter in (1), the weighted segment estimators are applied separately to each interval $[t_{k-1}, t_k]$. If the segment-level estimators use disjoint observations between thresholds and the errors are independent conditional on the design, then the segment-level estimators are independent conditional on the design. A finite weighted combination of segment-level estimators is therefore asymptotically normal, with variance equal to the corresponding weighted sum of segment-level variances.

6 Simulation Evidence

6.1 Single-segment evidence

We first examine the normalized single-segment problem. We compare three estimators of the smooth segment contrast

$$\delta = m(1^-) - m(0^+).$$

The first is a boundary-level estimator,

$$\hat{\delta}_{\text{bnd}} = \hat{m}(1^-) - \hat{m}(0^+),$$

computed using one-sided local linear regressions at 0^+ and 1^- with the triangular kernel $K(u) = \max\{0, 1 - |u|\}$. The second is the integrated-derivative estimator,

$$\hat{\delta}_{\text{int}} = \sum_{b=1}^B \Delta \hat{\beta}_1(c_b),$$

which estimates $m'(c_b)$ at bin midpoints using local quadratic regressions and aggregates the slope estimates by midpoint quadrature. The third is the inverse-variance combined estimator,

$$\hat{\delta}_{\text{comb}} = w_{\text{bnd}} \hat{\delta}_{\text{bnd}} + w_{\text{int}} \hat{\delta}_{\text{int}},$$

with weights computed from feasible variance estimates.

We partition $[0, 1]$ into B equal bins with $\Delta = 1/B$ and set $h_0 = \Delta/2$. The number of bins is chosen as $B(n) = \max\{4, \text{round}(1.5n^{1/4})\}$, so $B = 10$ when $n = 2000$. For bin b , we run a local quadratic regression of Y on $(1, X - c_b, (X - c_b)^2)$ with triangular weights on $|X - c_b| \leq h_0$, and take the slope coefficient $\hat{\beta}_1(c_b)$ as the estimate of $m'(c_b)$. The feasible variance estimator for the integrated estimator is based on

$$\hat{\Omega}^2 = 2C_K \sum_{b=1}^B \Delta \frac{\hat{\sigma}_b^2}{\hat{f}_b}, \quad \widehat{\text{SE}}_{\text{int}} = \sqrt{\hat{\Omega}^2 / (nh_0^2)},$$

where $C_K = \int K^*(u)^2 du$ is the derivative equivalent-kernel constant for local quadratic regression, $\hat{f}_b$ is a local density estimate based on bin counts, and $\hat{\sigma}_b^2$ is the local residual mean squared error from the derivative fit. For the boundary-level estimator, the plug-in variance uses one-sided equivalent-kernel constants together with local residual variances and boundary density estimates at 0^+ and 1^- .

Remark 1 (Implementation bandwidth). The rule $B(n) = \max\{4, \text{round}(1.5n^{1/4})\}$ is a feasible implementation rule used in the simulations, not an optimal bandwidth rule. Since $h_0 = \Delta/2 = 1/(2B)$, this choice gives $h_0 \asymp n^{-1/4}$, which satisfies the bandwidth restrictions in Assumption 5. Deriving an optimal band-

width selector for the integrated-derivative estimator, and for the combined estimator, is a separate problem. In applications, one can report sensitivity to the bin and bandwidth choice, or combine this rule with standard local-polynomial bandwidth choices for the boundary-level estimator.

We draw X_i either from a uniform distribution on $[0, 1]$ or from a truncated normal distribution $\mathcal{N}(0.5, 0.3^2)$ on $[0, 1]$. The latter design places less mass near the endpoints and therefore weakens the boundary-level estimator. Outcomes are generated as

$$Y_i = m(X_i) + \varepsilon_i,$$

with $\varepsilon_i \sim \mathcal{N}(0, \sigma^2(X_i))$. We consider both homoskedastic errors, $\sigma(x) = 1$, and heteroskedastic errors, $\sigma^2(x) = 1 + x$. We use $nsim = 500$ Monte Carlo replications and report bias, standard deviation, root mean squared error, and the empirical coverage of the feasible 95% Wald interval for the integrated estimator.

We consider two conditional mean functions:

$$m(x) = 2x + x^2, \quad m(x) = \frac{1 - \cos(\pi x)}{2}.$$

For the polynomial design, $\delta = 3$. For the cosine design, $\delta = 1$. The simulation code is available at this repository.

The combined estimator reduces RMSE relative to the boundary-level estimator in all 12 single-segment designs. The improvement is modest under regular uniform designs and larger under the truncated-normal designs, where fewer observations are available near the endpoints. This pattern is consistent with the information-fusion result: the integrated estimator is most useful when boundary information is weak because it provides an additional signal based on interior variation.

Scenario	Boundary			Integrated				Combined		
	Bias	SD	RMSE	Bias	SD	RMSE	Cov.95	Bias	SD	RMSE
<i>Panel A: Polynomial DGP $m(x) = 2x + x^2$</i>										
Uniform – Homo	0.003	0.340	0.340	−0.013	0.983	0.982	0.950	0.002	0.328	0.327
Uniform – Het	−0.001	0.415	0.415	−0.032	1.191	1.190	0.954	−0.005	0.406	0.405
Trunc. Normal (sd=0.3) – Homo	−0.009	0.491	0.491	−0.021	1.147	1.146	0.936	−0.013	0.476	0.476
Trunc. Normal (sd=0.3) – Het	−0.005	0.629	0.628	−0.094	1.429	1.430	0.926	−0.018	0.612	0.612
Uniform – Homo [n=1000, B=8]	0.009	0.409	0.409	−0.046	1.096	1.096	0.948	0.002	0.394	0.394
Uniform – Homo [n=4000, B=12]	0.007	0.231	0.231	0.005	0.813	0.812	0.950	0.007	0.226	0.226
<i>Panel B: Cosine DGP $m(x) = \frac{1 - \cos(\pi x)}{2}$</i>										
Uniform – Homo	0.000	0.303	0.303	−0.049	0.967	0.967	0.956	−0.003	0.296	0.295
Uniform – Het	−0.032	0.379	0.380	−0.085	1.217	1.218	0.958	−0.036	0.369	0.370
Trunc. Normal (sd=0.3) – Homo	0.059	0.637	0.639	0.039	1.114	1.114	0.942	0.055	0.607	0.609
Trunc. Normal (sd=0.3) – Het	−0.033	0.700	0.700	−0.109	1.342	1.345	0.932	−0.047	0.655	0.656
Uniform – Homo [n=1000, B=8]	0.021	0.368	0.368	0.026	1.063	1.062	0.958	0.021	0.360	0.360
Uniform – Homo [n=4000, B=12]	0.013	0.225	0.225	0.009	0.820	0.819	0.958	0.013	0.219	0.219

Notes: The baseline sample size is $n = 2000$ unless otherwise indicated. The table reports the feasible-weight implementation with $B(n) = \max\{4, \text{round}(1.5n^{1/4})\}$. Cov.95 is the empirical coverage of the feasible 95% Wald interval for the integrated-derivative estimator.

Table 1: Single-segment simulation results: integrated-derivative, boundary-level, and combined estimators

6.2 Ordered multi-threshold evidence

We simulate ordered multi-threshold RD designs with $K \in \{5, 10\}$ cutoffs at

$$t_k = \frac{k}{K+1}, \quad k = 1, \dots, K.$$

The target is the equal-weight average of the local discontinuities. Outcomes are generated from

$$Y_i = q(Z_i) + \sum_{k=1}^K \tau_k \mathbf{1}\{Z_i \geq t_k\} + \varepsilon_i,$$

where the jumps are heterogeneous but centered so that $K^{-1} \sum_k \tau_k = 0.6$. We consider two smooth baselines,

$$q(z) = 2z + z^2, \quad q(z) = \frac{1 - \cos(\pi z)}{2},$$

and draw Z_i either uniformly on $[0, 1]$ or from $\mathcal{N}(0.5, 0.3^2)$ truncated to $[0, 1]$. Errors are either homoskedastic, $\sigma(z) = 1$, or heteroskedastic, $\sigma^2(z) = 1 + z$. Each design uses 300 Monte Carlo replications.

We compare two estimators. The first estimates each cutoff-specific RD jump using one-sided local linear regressions and averages the estimates with equal weights. The second uses the decomposition in (1): the two outer end-point terms are estimated by one-sided local linear regressions, while each between-threshold segment contrast is estimated by the inverse-variance combination of its boundary-level and integrated-derivative estimators. Each segment is normalized to $[0, 1]$ before applying the integrated-derivative estimator, with $B(n) = \max\{4, \text{round}(1.5n^{1/4})\}$ bins.

Baseline	Design	K	Errors	Sep. RMSE	Sep. Cov.	Comb. RMSE	Comb. Cov.	Gain
Poly.	Uniform	5	Homo	0.160	0.940	0.149	0.947	6.40
Poly.	Uniform	10	Homo	0.109	0.950	0.109	0.933	0.27
Cos.	Uniform	5	Homo	0.150	0.963	0.148	0.963	1.19
Cos.	Uniform	10	Homo	0.102	0.950	0.101	0.930	1.04
Poly.	Trunc. normal	5	Homo	0.149	0.957	0.146	0.943	1.82
Poly.	Trunc. normal	10	Homo	0.113	0.940	0.109	0.930	3.70
Cos.	Trunc. normal	5	Homo	0.154	0.943	0.152	0.930	1.30
Cos.	Trunc. normal	10	Homo	0.104	0.957	0.104	0.947	0.24
Poly.	Uniform	5	Het.	0.194	0.930	0.191	0.920	1.62
Poly.	Uniform	10	Het.	0.128	0.947	0.129	0.940	-0.85
Cos.	Uniform	5	Het.	0.187	0.950	0.184	0.947	2.01
Cos.	Uniform	10	Het.	0.127	0.953	0.124	0.923	1.99
Poly.	Trunc. normal	5	Het.	0.195	0.933	0.191	0.923	2.23
Poly.	Trunc. normal	10	Het.	0.141	0.923	0.135	0.933	4.12
Cos.	Trunc. normal	5	Het.	0.188	0.933	0.184	0.947	2.08
Cos.	Trunc. normal	10	Het.	0.122	0.963	0.121	0.947	0.68

Notes: Sep. is the equal-weight average of cutoff-specific local linear RD estimates. Comb. is the decomposition estimator that combines boundary and integrated-derivative signals for each interior segment. Gain is the percentage RMSE reduction of Comb. relative to Sep.

Table 2: Multi-threshold simulation results

The combined decomposition estimator reduces RMSE relative to the separate RD benchmark in 15 of the 16 designs. Coverage of the feasible 95% intervals remains close to nominal for both estimators. The gains are moderate under regular designs and larger when boundary information is weaker, consistent with the information-fusion result.

7 Discussion and Conclusion

This paper studies estimation of smooth segment contrasts that arise in ordered multi-threshold regression discontinuity designs. We show that aggre-

gate RD estimands can be decomposed into two components: local discontinuities at thresholds and smooth between-threshold contrasts. While standard RD methods identify the former using observations near the cutoffs, the latter depend on differences of endpoint limits across adjacent segments, such as $m(1^-) - m(0^+)$ after normalization.

The main contribution is to provide a second estimator for these smooth segment contrasts. The integrated-derivative estimator exploits the identity

$$m(1^-) - m(0^+) = \int_0^1 m'(u) du,$$

and uses interior observations to estimate the derivative and aggregate it over the segment. The estimator admits a feasible central limit theorem with an explicit variance expression that depends on the design density and conditional variance over the segment.

A key result is that the integrated-derivative estimator and the conventional boundary-level estimator are asymptotically independent. The independence follows from the geometry of their weighting schemes: the boundary estimator concentrates on shrinking neighborhoods of 0^+ and 1^- , while the integrated estimator distributes weight over the interior of the segment. The overlap between these effective supports vanishes as the bandwidth shrinks. This property yields a simple and implementable inverse-variance combination rule, which uses both signals for the same parameter.

The implications are design-dependent. When observations are sparse near 0^+ or 1^- , or when the conditional variance is high at the segment endpoints, the interior estimator provides complementary information. In such cases, the combined estimator can deliver non-negligible improvements in precision. The simulation results confirm this pattern: gains are modest under uniform designs and more pronounced under irregular designs with weak boundary information.

The approach relies on smoothness of the conditional mean within each segment. This assumption is necessary for representing the segment contrast

as an integral of the derivative and for consistent estimation of $m'(\cdot)$. If there are latent discontinuities or structural breaks between observed thresholds, the integrated representation is invalid and the estimator is not appropriate. In addition, while asymptotic independence guarantees that the two estimators can be combined without accounting for covariance at first order, finite-sample performance depends on bandwidth choices and local sample sizes, and the interior estimator alone can be noisy.

Finally, the analysis assumes known and ordered thresholds. Extensions to settings with endogenous or estimated cutoffs, as well as to designs with multidimensional running variables or interacting thresholds, remain open. More broadly, the decomposition in (1) suggests that multi-threshold RD designs contain distinct sources of information—boundary discontinuities and interior smooth variation—and that efficient estimation requires combining both in a manner that respects their different identification content.

References

- Bertanha, M. (2020, September). Regression discontinuity design with many thresholds. *Journal of Econometrics* 218(1), 216–241.
- Calonico, S., M. D. Cattaneo, and M. H. Farrell (2020). Optimal bandwidth choice for robust bias-corrected inference in regression discontinuity designs. *The Econometrics Journal* 23(2), 192–210.
- Calonico, S., M. D. Cattaneo, and R. Titiunik (2014). Robust nonparametric confidence intervals for regression-discontinuity designs. 82(6), 2295–2326. Publisher: Wiley Online Library.
- Cattaneo, M. D., L. Keele, R. Titiunik, and G. Vazquez-Bare (2016). Interpreting regression discontinuity designs with multiple cutoffs. 78(4), 1229–1248. Publisher: The University of Chicago Press.
- Fan, J. and Q. Yao (2005). *Nonlinear Time Series: Nonparametric and Parametric Methods* (1 ed.). Springer Series in Statistics. New York: Springer New York, NY. Springer Book Archive.
- Hahn, J., P. Todd, and W. Van der Klaauw (2001). Identification and estimation of treatment effects with a regression-discontinuity design. *Econometrica* 69(1), 201–209.

Appendix

A Proofs

A.1 Proof of Theorem 1

Write

$$\hat{\beta}_{\text{avg}} - \Gamma = 2h \sum_{b=1}^B (\hat{\beta}_1(c_b) - m'(c_b)) + \left\{ 2h \sum_{b=1}^B m'(c_b) - \int_0^1 m'(u) du \right\}.$$

By Lemma 3,

$$2h \sum_{b=1}^B (\hat{\beta}_1(c_b) - m'(c_b)) = \frac{h^p}{(p+1)!} \left(\int_{-1}^1 u^{p+1} K^*(u) du \right) [m^{(p)}(1^-) - m^{(p)}(0^+)] + o(h^p).$$

By Lemma 2, applied to $g = m'$,

$$2h \sum_{b=1}^B m'(c_b) - \int_0^1 m'(u) du = -\frac{h^2}{6} [m''(1^-) - m''(0^+)] + O(h^4).$$

Thus, if $p > 2$, the leading bias is

$$\text{Bias} = -\frac{h^2}{6} [m''(1^-) - m''(0^+)].$$

If $p = 2$, the local-polynomial and midpoint terms both contribute at order h^2 , giving

$$\text{Bias} = \frac{h^2}{6} \left(\int_{-1}^1 u^3 K^*(u) du - 1 \right) [m''(1^-) - m''(0^+)].$$

For the stochastic term, write

$$\hat{\beta}_{\text{avg}} - \mathbb{E}[\hat{\beta}_{\text{avg}} \mid X] = \sum_{i=1}^N w_{N,i} \varepsilon_i.$$

By Lemma 4,

$$\text{Var} \left(\sum_{i=1}^N w_{N,i} \varepsilon_i \mid X \right) = \frac{2}{Nh^2} \left(\int_{-1}^1 K^{*2}(u) du \right) \left(\int_0^1 \frac{\sigma^2(u)}{f(u)} du \right) + o \left(\frac{1}{Nh^2} \right).$$

Let

$$\Omega^2 = 2 \left(\int_{-1}^1 K^{*2}(u) du \right) \left(\int_0^1 \frac{\sigma^2(u)}{f(u)} du \right).$$

Lemma 1 gives

$$\frac{\sqrt{Nh^2}}{\Omega} (\hat{\beta}_{\text{avg}} - \mathbb{E}[\hat{\beta}_{\text{avg}} \mid X]) \xrightarrow{d} \mathcal{N}(0, 1).$$

The remaining deterministic terms are negligible under Assumption 5. Combining the bias expansion, the conditional CLT, and Slutsky's lemma proves the result. $\square$

A.2 Proof of Theorem 2

Write

$$S_\beta = \hat{\beta}_{\text{avg}} - \mathbb{E}[\hat{\beta}_{\text{avg}} \mid X] = \sum_{i=1}^N w_i \varepsilon_i$$

and

$$S_\delta = \hat{\delta}_{\text{levels}} - \mathbb{E}[\hat{\delta}_{\text{levels}} \mid X] = \sum_{i=1}^N v_i \varepsilon_i.$$

Conditional on X ,

$$\text{Cov}(S_\beta, S_\delta \mid X) = \sum_{i=1}^N w_i v_i \sigma^2(X_i).$$

By construction, $v_i \neq 0$ only if

$$X_i \in \mathcal{O}_h := [0, A_0 h] \cup [1 - A_1 h, 1].$$

Since f has finite one-sided limits at the endpoints and no atoms,

$$\mathbb{P}(X_i \in \mathcal{O}_h) = A_0 h f(0+) + A_1 h f(1-) + o(h).$$

Hence,

$$\#\{i : X_i \in \mathcal{O}_h\} = N\{A_0f(0+) + A_1f(1-)\}h + o_p(Nh).$$

Using

$$\max_i |w_i| \leq \frac{C}{Nh}, \quad \max_i |v_i| \leq \frac{C}{Nh},$$

and boundedness of $\sigma^2(\cdot)$,

$$|\text{Cov}(S_\beta, S_\delta \mid X)| \leq C\#\{i : X_i \in \mathcal{O}_h\} \frac{1}{Nh} \frac{1}{Nh} = O_p\left(\frac{A_0f(0+) + A_1f(1-)}{Nh}\right).$$

Moreover,

$$\text{Var}(S_\beta \mid X) = \Theta_p\left(\frac{1}{Nh^2}\right), \quad \text{Var}(S_\delta \mid X) = \Theta_p\left(\frac{1}{Nh}\right).$$

Therefore,

$$\text{Corr}(S_\beta, S_\delta \mid X) = O_p\left(\{A_0f(0+) + A_1f(1-)\}h^{1/2}\right) \rightarrow 0.$$

Together with the marginal central limit theorems for $\hat{\beta}_{\text{avg}}$ and $\hat{\delta}_{\text{levels}}$, the vanishing correlation implies joint asymptotic normality with diagonal limiting covariance. $\square$

A.3 Proof of Theorem 3

Using (5) and (6), write

$$\begin{aligned} \hat{\delta}_w - \delta_w = & \underbrace{2h \sum_b \pi(c_b) (\hat{\beta}_1(c_b) - m'(c_b))}_A + \underbrace{\left\{ 2h \sum_b \pi(c_b) m'(c_b) - \int_0^1 \pi(u) m'(u) du \right\}}_B \\ & + \underbrace{\int_0^1 \pi'(u) (\hat{m}(u) - m(u)) du}_C. \end{aligned}$$

Term A is handled by Lemma 3 with the additional factor $\pi(c_b)$. Term B is the midpoint quadrature error applied to $g(u) = \pi(u)m'(u)$, giving

$$B = -\frac{h^2}{6} [(\pi m')'(1^-) - (\pi m')'(0^+)] + O(h^4).$$

Term C is $O_p(N^{-1/2})$ by assumption, and hence

$$\sqrt{Nh^2} C = o_p(1).$$

Combining the local-polynomial and midpoint bias terms gives the stated expression for Bias_π . The variance calculation follows from Lemma 4 with the additional multiplicative factor $\pi(c_b)$, yielding

$$2 \int_{-1}^1 K^{*2}(u) du \int_0^1 \frac{\pi^2(u)\sigma^2(u)}{f(u)} du.$$

The conditional central limit theorem follows from Lemma 1 applied to the corresponding weighted linear representation. $\square$

Remark 2 (Data-chosen π is first-order negligible). If $\pi(\cdot)$ is selected from the quadratic family by the constraints $\pi(1^-) = \omega_1$, $\pi(0^+) = \omega_0$, and the sample orthogonality condition $\int_0^1 \pi'(u)\hat{m}(u) du = 0$, then the second term of (6) is zero by construction. Moreover, letting π^* denote the population solution with $\int_0^1 \pi^{*'}(u)m(u) du = 0$, one has $\|\hat{\pi} - \pi^*\|_\infty = O_p(N^{-1/2})$, and

$$\sqrt{Nh^2} (\hat{\delta}_w(\hat{\pi}) - \hat{\delta}_w(\pi^*)) = o_p(1).$$

Thus, the stochastic choice of π does not affect the first-order limit law.

Remark 3 (Quadrature order and bandwidth restrictions). The midpoint aggregation induces an $O(h^2)$ quadrature error for $\int_0^1 m'(u) du$. If a composite quadrature rule of global order $q \geq 2$ is used instead, the quadrature error becomes $O(h^q)$. Since the aggregate local-polynomial bias is $O(h^p)$, the deterministic remainder is $O(h^{\min\{p,q\}})$. Under the normalization $\sqrt{Nh^2}$, it suffices

that

$$\sqrt{N}h^{1+\min\{p,q\}} \rightarrow 0.$$

Thus, the bandwidth restriction in Theorem 1 can be relaxed with higher-order quadrature rules.

A.4 Auxiliary lemmas

Lemma 1 (Convergence in Distribution). *Under Assumptions 1–6, for each sample size N , consider the triangular array*

$$\{w_{N,i}\varepsilon_i : i = 1, \dots, N\}.$$

Define

$$T_N = \sum_{i=1}^N w_{N,i}\varepsilon_i, \quad v_N^2 = \text{Var}(T_N \mid X) = \sum_{i=1}^N w_{N,i}^2 \sigma^2(X_i).$$

Suppose there exists a deterministic sequence $\alpha_N > 0$ such that

$$\alpha_N^2 v_N^2 \xrightarrow{p} \Omega^2 > 0.$$

Then

$$\frac{\alpha_N T_N}{\Omega} \xrightarrow{d} \mathcal{N}(0, 1).$$

Proof. The effective weights satisfy

$$\max_i |w_{N,i}| = O_p\left(\frac{1}{Nh}\right), \quad \sum_{i=1}^N w_{N,i}^2 = O_p\left(\frac{1}{Nh^2}\right).$$

Conditional on X , the summands $w_{N,i}\varepsilon_i$ are independent and mean zero. By Assumption 6,

$$\sum_{i=1}^N \mathbb{E} \left[|w_{N,i}\varepsilon_i|^{2+\delta} \mid X \right] \leq C \sum_{i=1}^N |w_{N,i}|^{2+\delta} = O_p\left(\frac{1}{N^{1+\delta}h^{2+\delta}}\right).$$

Moreover,

$$(v_N^2)^{1+\delta/2} = O_p\left(\frac{1}{N^{1+\delta/2}h^{2+\delta}}\right).$$

Therefore the Lyapunov ratio is $O_p(N^{-\delta/2}) = o_p(1)$. The triangular-array Lyapunov CLT gives

$$\frac{T_N}{v_N} \xrightarrow{d} \mathcal{N}(0, 1).$$

Since $\alpha_N^2 v_N^2 \rightarrow_p \Omega^2$, Slutsky's lemma proves the result. $\square$

Lemma 2 (Midpoint Rule Approximation Error). *Let $g \in C^4([0, 1])$. Partition $[0, 1]$ into B subintervals of length $2h = 1/B$, with midpoints $c_b = (2b - 1)h$. Define*

$$E = \int_0^1 g(u) du - 2h \sum_{b=1}^B g(c_b).$$

Then

$$E = \frac{h^2}{6} \{g'(1^-) - g'(0^+)\} + O(h^4).$$

Proof. On each interval $[a_b, a_b + 2h]$, with midpoint $c_b = a_b + h$, Taylor expansion gives

$$\int_{a_b}^{a_b+2h} g(u) du = 2h g(c_b) + \frac{1}{3} h^3 g''(c_b) + O(h^5).$$

Summing over b yields

$$E = \frac{1}{3} h^3 \sum_{b=1}^B g''(c_b) + O(h^4).$$

Since

$$2h \sum_{b=1}^B g''(c_b) = \int_0^1 g''(u) du + O(h^2) = g'(1^-) - g'(0^+) + O(h^2),$$

the result follows. $\square$

Lemma 3 (Aggregate Bias of Local Polynomial Estimator). *Under Assumptions 1–6, let $\hat{\beta}_1(c_b)$ denote the degree- p local-polynomial estimator of $m'(c_b)$. Partition $[0, 1]$*

into B subintervals of width $2h = 1/B$, with midpoints $c_b = (2b - 1)h$, and define

$$R = 2h \sum_{b=1}^B \hat{\beta}_1(c_b) - 2h \sum_{b=1}^B m'(c_b).$$

Let $\mathcal{B}_N(h) = \mathbb{E}[R \mid X]$. Then

$$\mathcal{B}_N(h) = \frac{h^p}{(p+1)!} \left(\int_{-1}^1 u^{p+1} K^*(u) du \right) \{m^{(p)}(1^-) - m^{(p)}(0^+)\} + o(h^p).$$

Proof. For each interior midpoint c_b , the standard equivalent-kernel expansion gives

$$\mathbb{E}[\hat{\beta}_1(c_b) \mid X] - m'(c_b) = \frac{h^p}{(p+1)!} m^{(p+1)}(c_b) \int_{-1}^1 u^{p+1} K^*(u) du + h^p \rho_b(h),$$

where $\sup_b |\rho_b(h)| = o(1)$. Multiplying by $2h$ and summing gives

$$\mathcal{B}_N(h) = \frac{2h^{p+1}}{(p+1)!} \left(\int_{-1}^1 u^{p+1} K^*(u) du \right) \sum_{b=1}^B m^{(p+1)}(c_b) + o(h^p).$$

Since

$$2h \sum_{b=1}^B m^{(p+1)}(c_b) \rightarrow \int_0^1 m^{(p+1)}(u) du = m^{(p)}(1^-) - m^{(p)}(0^+),$$

the result follows. $\square$

Lemma 4 (Variance of the Integrated-Derivative Estimator). *Under Assumptions 1–6,*

$$\text{Var}(\hat{\beta}_{\text{avg}} \mid X) = \frac{2}{Nh^2} \int_{-1}^1 K^{*2}(u) du \int_0^1 \frac{\sigma^2(u)}{f(u)} du + o\left(\frac{1}{Nh^2}\right).$$

Proof. Because the kernel is supported on $[-1, 1]$ and the bins have width $2h$, the observations used in different midpoint regressions are disjoint. Conditional on X ,

$$\text{Var}(\hat{\beta}_{\text{avg}} \mid X) = 4h^2 \sum_{b=1}^B \text{Var}(\hat{\beta}_1(c_b) \mid X).$$

The local-polynomial variance expansion gives, uniformly over b ,

$$\text{Var}(\hat{\beta}_1(c_b) \mid X) = \frac{\sigma^2(c_b)}{Nh^3 f(c_b)} \int_{-1}^1 K^{*2}(u) du + o\left(\frac{1}{Nh^3}\right).$$

Therefore,

$$\text{Var}(\hat{\beta}_{\text{avg}} \mid X) = \frac{4}{Nh} \left(\int_{-1}^1 K^{*2}(u) du \right) \sum_{b=1}^B \frac{\sigma^2(c_b)}{f(c_b)} + o\left(\frac{1}{Nh^2}\right).$$

Since

$$2h \sum_{b=1}^B \frac{\sigma^2(c_b)}{f(c_b)} \rightarrow \int_0^1 \frac{\sigma^2(u)}{f(u)} du,$$

the result follows. □